ACI Notes – Part 7

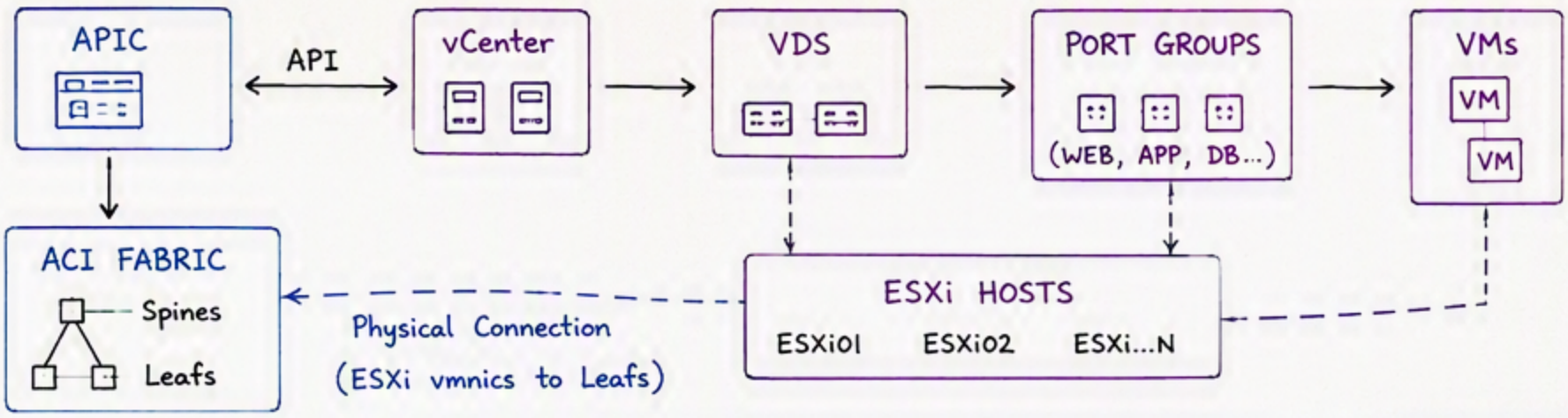
VMM DOMAIN INTEGRATION (VMWARE)

ACI integrates with VMware vCenter to automatically create and manage networking objects such as VDS, Port Groups for EPGs.

KEY BENEFITS

- ✓ Automation
- ✓ VM Mobility (vMotion)
- ✓ Policy Follows VM
- ✓ No manual VLAN/PortGroup
- ✓ Micro-Segmentation

1. BIG PICTURE



2. CORE COMPONENTS

① VMM DOMAIN : Container for vCenter, VDS, VLAN Pool, AEP, EPGs.

② vCENTER : APIC connects using credentials.

③ VLAN POOL : Dynamic VLAN allocation for Port Groups (e.g. 3000-3999).

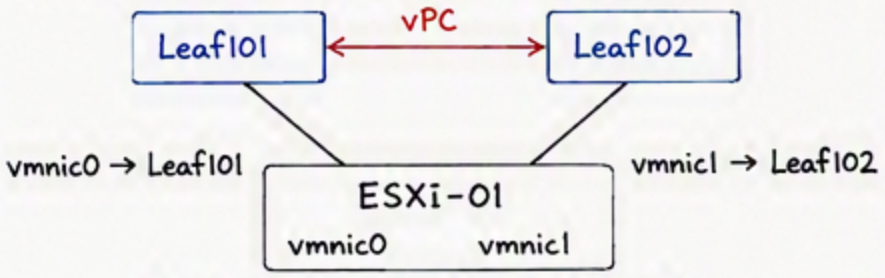
④ AEP (Attachable Access Entity Profile): Bridges Physical Fabric and VMM Domain.

5) VDS (Distributed Switch): APIC creates and manages Port Groups in vCenter.

3. WORKFLOW (EPG to VM)

1) Create EPG (e.g. EPG-WEB)
↓
2) Associate EPG to VMM Domain
↓
3) APIC automatically creates Port Group (e.g. PG-WEB) on VDS in vCenter.
↓
4) VMware Admin attaches VM NIC to PG-WEB.
↓
5) VM becomes member of EPG-WEB.
↓
6) Traffic flows through ESXi vmnic to ACI Leafs.

4. PHYSICAL TOPOLOGY EXAMPLE



(Redundancy + Active/Active)

5. VMM DOMAIN DISCOVERY

- APIC learns vCenter inventory (ESXi hosts, VMs, PortGroups).
- Learns ESXi <-> Leaf adjacency via LLDP/CDP.

Verify:

show vmm domain
show vmm controller

} ESXi hosts should be visible

6. DYNAMIC vs STATIC BINDING

STATIC BINDING	VMM DYNAMIC BINDING
• For Physical Servers	• For VMware VMs
• Manual interface mapping	• Auto Port Group creation
• Manual VLAN	• Dynamic VLAN from Pool
• Fixed location	• Supports vMotion
• No support for vMotion	• Policy follows VM

✦ Always prefer VMM Dynamic Binding for VMware.

7. TROUBLESHOOTING CHECKLIST

① Is VMM Domain Operational ? (APIC <-> vCenter)
② Are ESXi Hosts visible under VMM Domain ?
③ Is EPG associated to VMM Domain ?
④ Was Port Group created in vCenter ?
⑤ Is VM attached to correct Port Group ?
⑥ Is AEP configured and associated correctly ?
⑦ Is VLAN Pool available and in range ?
⑧ Can endpoint be learned ?
  • show endpoint ip <IP>

8. IMPORTANT NOTES

- ✓ Port Group belongs to VDS, not to individual ESXi host.
- ✓ One VMM Domain can support many ESXi hosts.
- ✓ vMotion does not require any ACI reconfiguration.
- ✓ Tag Collection must be enabled for ESG/VM Tag classification.
- ✓ Allow Micro-Segmentation must be enabled on VMM Association when using VMware tags/names for classification.

9. MEMORY AID

EPG (APIC) → Port Group (Created by APIC in vCenter/VDS) → VM (Attached by VM Admin)

VM joins EPG through Port Group.